

Evaluating the Welfare Impacts of EV Charging Station Subsidies: An Analysis of the NEVI Program in Texas

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October 2026

Abstract

Public support for electric vehicles has long focused on purchase subsidies, and federal funding is now shifting toward the charging network, yet the welfare return on charging subsidies is rarely measured. I evaluate the first round of the National Electric Vehicle Infrastructure (NEVI) program in Texas, which commits \$31.8 million to 50 new charging stations along interstate corridors. Using ZIP-code-level registrations of each EV model from 2017 to 2023, I estimate a random-coefficients demand model, disciplined by micro moments on EV buyers' income and homeownership, in which local charging availability enters the value of each model. Most models record no sales in most ZIP-code-quarters; rather than dropping these zeros or aggregating to the state level, I bound demand with the moment inequalities of Gandhi et al. (2023) and construct confidence sets following Chernozhukov et al. (2007). Simulating a year with and without the 50 stations, I find expected annual welfare gains, summing consumer surplus, vehicle producer profits, and emission reductions, of about \$15.2 million against \$31.8 million in fixed costs, implying a payback period of roughly two years.

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